

Para Bank Automation Test Plan

Selenium + TestNG + Maven | Page Object Model Framework | CI/CD Pipeline

Created By: Tapabrata Sikder

Batch: SDET-B3

Date: 16/06/2026

GitHub: [GitHub Link](#)

Video Link: [Project Video](#)



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1. Introduction

Banking applications process highly sensitive financial information and therefore demand extensive, repeatable validation to ensure reliability, accuracy, and stability. Manual testing of routine banking workflows — registration, login, fund transfers, bill payments — is time-consuming and prone to human error, which makes automation a natural fit for this domain.

This project documents a Selenium + TestNG + Maven hybrid automation framework, built around the Page Object Model, to validate end-to-end banking workflows on the ParaBank application. The framework covers seven core functional modules — Registration, Login, Account Management, Fund Transfer, Bill Payment, and Logout — alongside an integrated negative login scenario, all executed within a single unified test flow. In the current execution cycle, all 8 test cases achieved a 100% pass rate.

Beyond functional validation, the framework integrates Jenkins CI/CD to ensure continuous feedback on every GitHub push, and follows a professional Git branching workflow for feature development. Six defects of High and Medium severity were identified during testing and logged in JIRA, reflecting proactive quality monitoring throughout the project lifecycle.

2. Project Objectives

- Automate critical banking workflows end-to-end
- Implement a reusable, maintainable Page Object Model framework
- Perform Data-Driven Testing using Apache POI and Excel
- Validate positive and negative business scenarios within a single unified flow
- Generate detailed execution reports (TestNG Emailable & Suite Reports)
- Support CI/CD execution through Jenkins
- Follow a professional Git branching workflow for feature development
- Identify, validate, and document critical application defects via JIRA

3. Test Environment Details

The automation suite was executed in the following environment. Tests were run on a Windows 11 (64-bit) machine using Google Chrome (v125) as the primary browser, managed via WebDriverManager which automatically resolves driver binaries at runtime. The target application is the publicly hosted ParaBank demo instance at <https://parabank.parasoft.com/parabank/index.htm>, treated as a staging/QA environment. No tests were executed against a local or production instance. Java 17 (JDK) was used as the runtime, paired with Maven 3.9.x for build and dependency management. All tests were executed in a single-threaded, sequential mode to preserve data dependencies across the end-to-end flow.

4. Tools and Technologies

Category	Technology
Language	Java
Automation Tool	Selenium WebDriver + WebDriverManager
Test Framework	TestNG
Build Management	Maven
Design Pattern	Page Object Model (POM)
Data Handling	Data-driven testing via Apache POI (Excel)
Reports	TestNG Emailable Report, Suite Report
CI/CD	Jenkins
Version Control	Git / GitHub
Branching Model (Git)	Feature branch workflow — negative-tests branch created for negative scenario development, merged to master after validation

5. Test Data Strategy

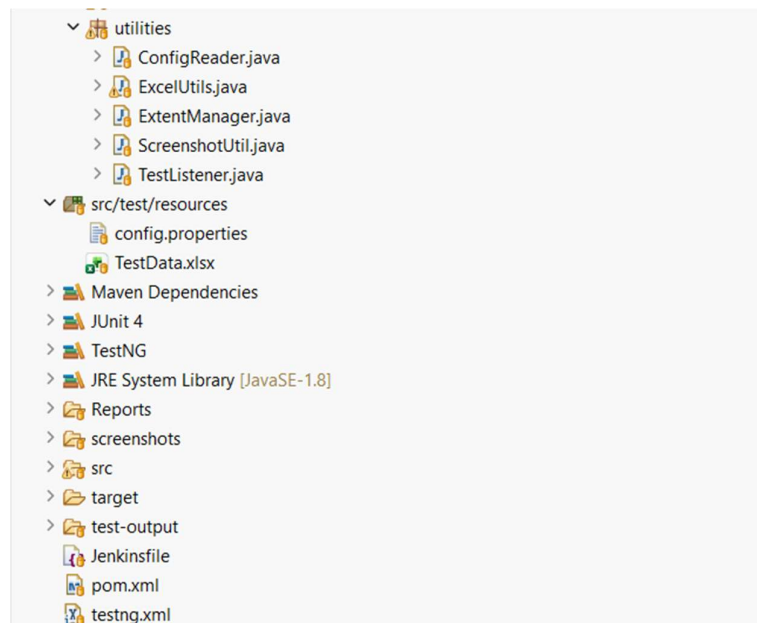
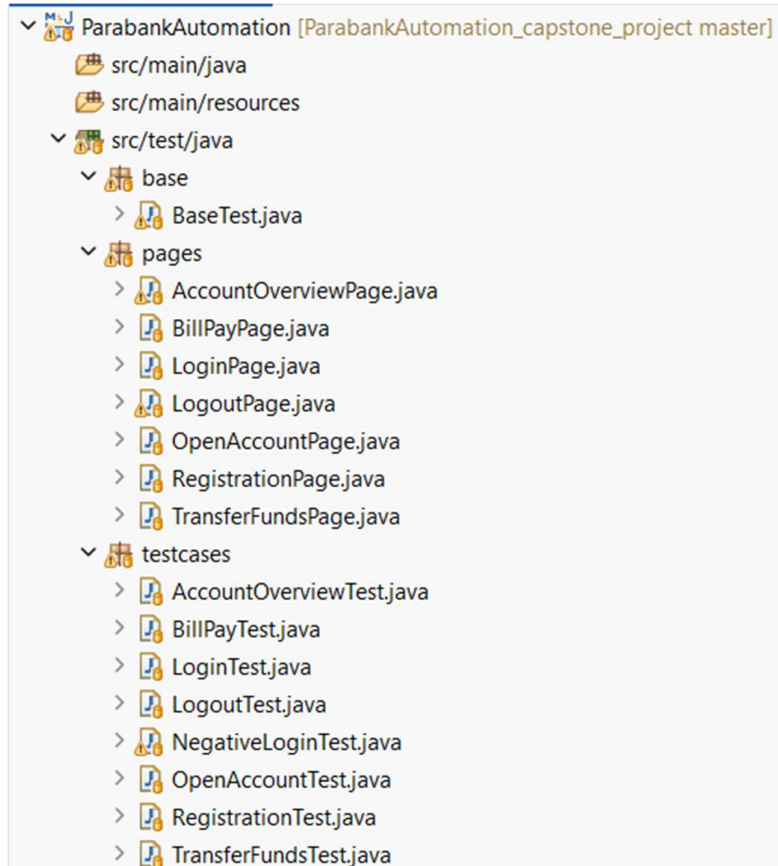
Test data for the automation suite is managed externally via an Apache POI-driven Excel file (TestData.xlsx) located in the src/test/resources/data directory of the project. This approach decouples test logic from test data, enabling easy data updates without modifying code. The Excel sheet contains the following fields for the registration and login flows: First Name, Last Name, Address, City, State, Zip Code, Phone Number, SSN, Username, and Password.

Sheet Name	Purpose	Used By
Sheet1	Valid registration data	RegistrationTest
Sheet2	Valid login credentials	LoginTest
Sheet3	Invalid login credentials for negative test scenarios	NegativeLoginTest

6. Framework Architecture

- **base/**: Contains BaseTest.java, which handles core configurations like browser initialization and driver teardown.
- **pages/**: Implements the Page Object Model (POM). Each class represents a specific web page (e.g., LoginPage.java, TransferFundsPage.java), separating UI locators from test logic.
- **testcases/**: Contains the executable TestNG classes that define the test scenarios.
- **utilities/**: Houses reusable helper methods for Excel data handling, configuration reading, and explicit waits.
- **test-output/ & screenshots/**: Stores generated reports and failure artifacts.

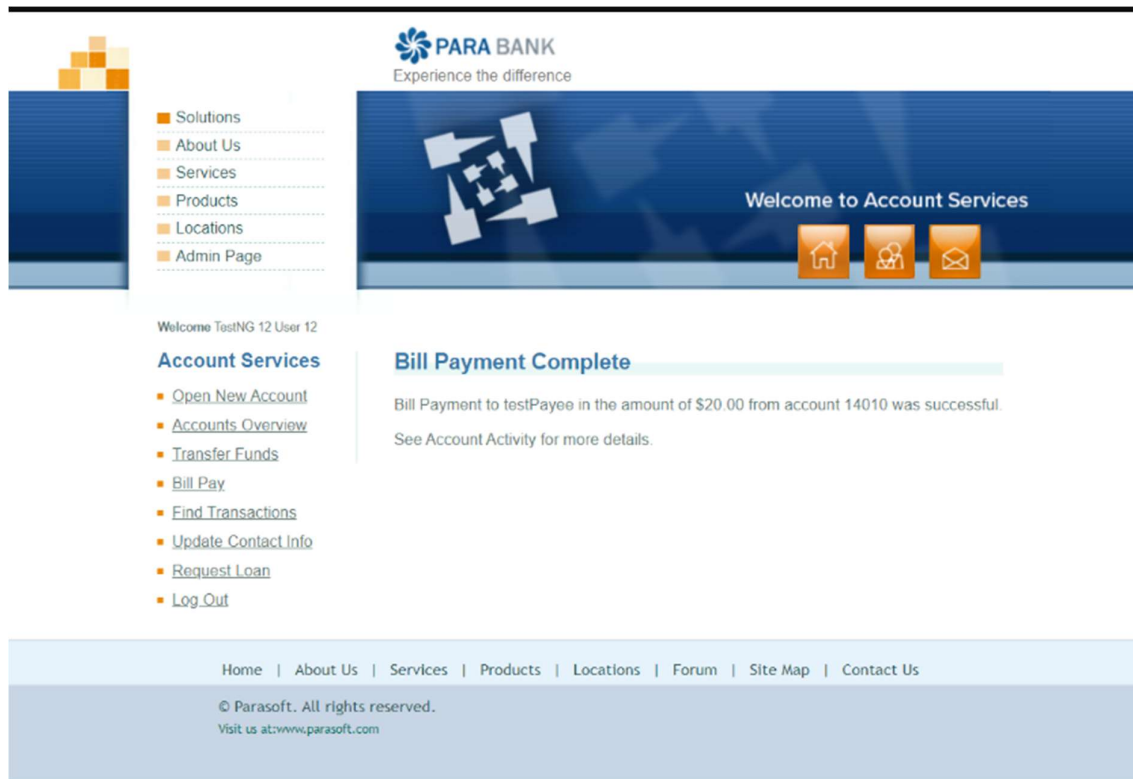
Below is the Structure of the project:



7. Automated Test Flow

The framework is designed to execute a comprehensive end-to-end banking workflow. Each test case utilizes the **Page Object Model (POM)** to interact with web elements, ensuring maintainability and reusability. The automated sequence is as follows:

- I. **User Registration:** The framework initiates by launching the Parabank application and performing a new user registration (e.g., testnguser04). It submits user credentials from excel file and verifies the successful creation of a new account profile.
- II. **Login:** Upon successful registration, the framework automates the login process using the newly created credentials, establishing a session to perform further banking transactions.
- III. **Account Management:**
 - **Open Account:** The framework navigates to the account opening module to create an additional savings/checking account. It captures and stores the newly generated **Account ID**.
 - **Account Overview:** The framework retrieves data from the account summary page. It validates the existence of both the primary and newly opened accounts and records their respective starting balances.
- IV. **Fund Transfer:** The framework performs a transaction by transferring a specified amount between the primary account (18006) and the newly opened account. It then validates the transaction success and confirms the updated account balances.
- V. **Bill Payment:** The framework executes a bill payment operation. It records the balance before and after the payment to ensure the transaction logic is accurate, and it automatically saves a **screenshot** of the success confirmation page for audit purposes.



- VI. **Logout:** Finally, the framework navigates to the logout portal to terminate the session securely. If the logout link is not detected (indicating the session has already ended), the framework gracefully handles the state to ensure the test suite finishes without errors.

Each test case in the framework performs explicit assertions using TestNG's Assert class to validate expected outcomes. The following table summarises the key assertions applied per module:

Module	Assertion Point	Validation Method
Registration	Welcome message contains registered username	Assert.assertTrue(welcomeText.contains(usrname))
Login	Post-login page title / account overview visible	Assert.assertEquals(pageTitle, expected)

Open Account	New Account ID is non-null and numeric	Assert.assertNotNull(accountId)
Account Overview	Both accounts appear in the account list	Assert.assertTrue(accountList.contains(accountId))
Fund Transfer	Confirmation message displayed; balances updated correctly	Assert.assertTrue(confirmMsg.isDisplayed())
Bill Payment	Payment confirmation page loaded; balance reduced by payment amount	Assert.assertEquals(newBalance, prevBalance - payAmount)
Logout	Session terminated; login page re-displayed	Assert.assertTrue(loginPage.isDisplayed())
Negative Login	Error message displayed; user not redirected	Assert.assertTrue(loginPage.isLoginErrorDisplayed())

8. Test Execution Summary

Test Case	Description	Status
RegistrationTest	Validates new user registration	PASS
LoginTest	Validates login with valid credentials	PASS
NegativeLoginTest	Validate login with invalid credentials	PASS
OpenNewAccountTest	Validates new account is opened	PASS
AccountOverviewTest	Validate Account overview for primary and new account	PASS
TransferFundsTest	Validates fund transfer workflow	PASS

BillPayTest	Validates bill payment functionality	PASS
LogoutTest	Validate logout functionality	PASS

Metric	Value
Total Test Cases Executed	8
Passed	8
Failed	0
Skipped	0
Overall Result	PASS

```
=====
ParaBank Suite
Total tests run: 8, Passes: 8, Failures: 0, Skips: 0
=====
```

9. Planned Test Scenarios

I. Implemented

The automation framework covers both positive (happy-path) and negative test scenarios within a single unified end-to-end flow. Negative test coverage has been integrated into the same execution suite rather than maintained as a separate test run. A dedicated NegativeLoginTest.java class has been created under the testcases/ package and is executed as part of the main TestNG suite alongside the positive test cases. This test uses a TestNG @DataProvider to feed invalid credentials from a separate Excel sheet (Sheet3 in TestData.xlsx) and asserts that the application correctly displays an error message on failed login attempts, verifying that the isLoginErrorDisplayed() method on LoginPage returns true. The test confirms the user is not redirected to the account overview page on invalid credentials.

II. Planned

The following additional negative and edge case scenarios have been identified for the next development phase and will be incorporated into the same unified test flow, continuing the approach of validating both positive and negative behaviours within a single suite execution:

- **Duplicate Username Registration:** Attempt to register with an already existing username and assert that the system rejects it with a clear validation message.
- **Empty Form Submissions:** Submit the registration and bill payment forms with empty required fields and verify that field-level validation errors are triggered.
- **Negative Amount in Fund Transfer (PAC-4):** Enter a negative value in the amount field and assert that the system rejects it — directly linked to the confirmed defect PAC-4.

These scenarios will be added to a dedicated NegativeTests.java class under the testcases/ package, following the existing POM structure. Test data for negative scenarios will be maintained in a separate sheet (Negative Data) within the existing TestData.xlsx file.

10. Test Case IDs & Requirements Traceability

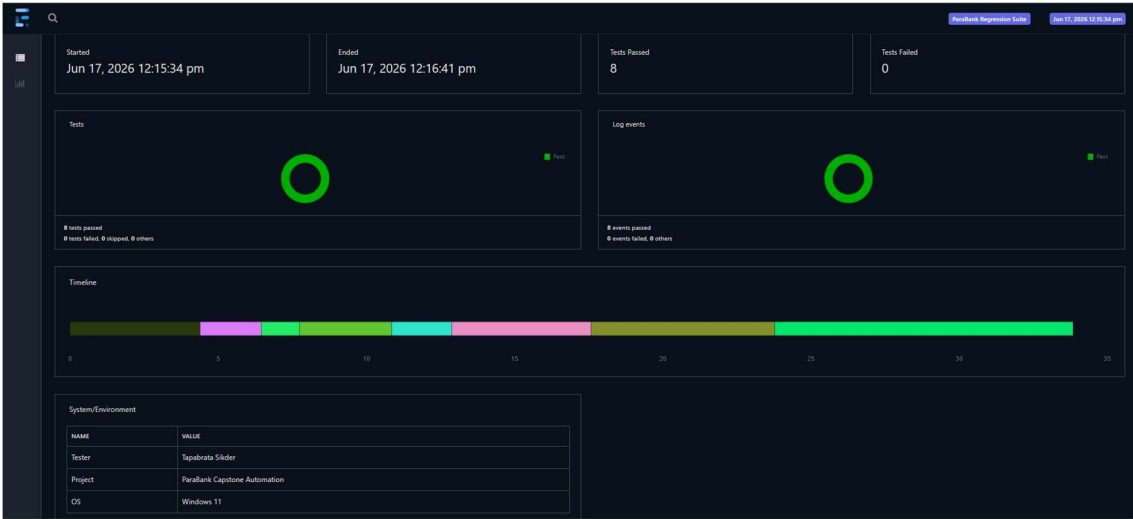
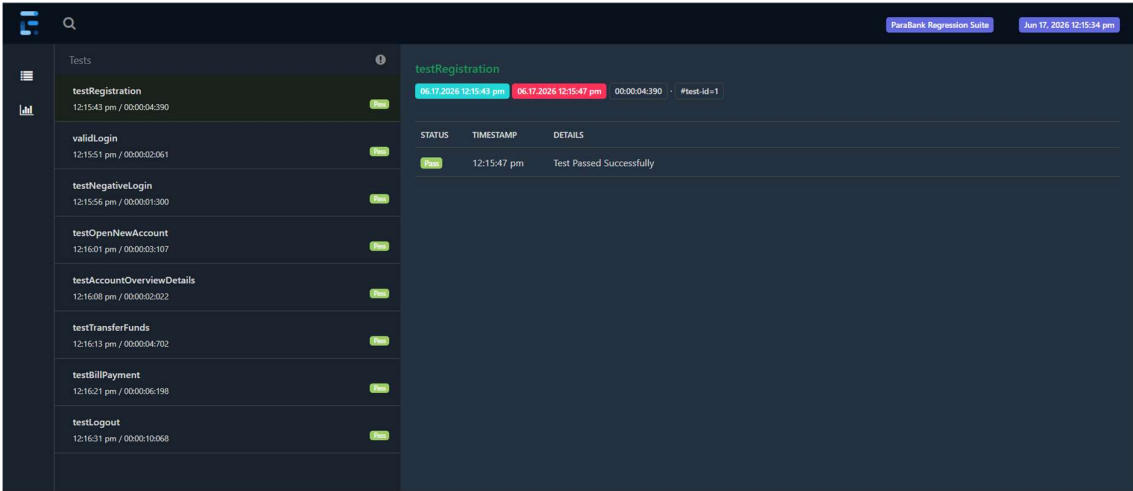
The table below maps each automated test case to its corresponding module and the related JIRA defects or requirement references. This traceability matrix ensures that all in-scope features have test coverage and provides a clear audit trail linking test execution results to functional requirements.

Test Case ID	Test Class	Module / Feature	Status	Linked JIRA
TC-001	RegistrationTest	User Registration	PASS	PAC-1
TC-002	LoginTest	Login (Valid)	PASS	—
TC-003	NegativeLoginTest	Login (Invalid Credentials)	PASS	PAC-1
TC-004	OpenNewAccountTest	Account Management	PASS	—

TC-005	AccountOverviewTest	Account Overview	PASS	PAC-6
TC-006	TransferFundsTest	Fund Transfer	PASS	PAC-2, PAC-4
TC-007	BillPayTest	Bill Payment	PASS	—
TC-008	LogoutTest	Logout	PASS	—

11. Test Reports

I. Extent Report



II. Emailable Report

Test	# Passed	# Skipped	# Retried	# Failed	Time (ms)	Included Groups	Excluded Groups
ParaBank Suite							
Regression Tests	8	0	0	0	55,630		

Class	Method	Start	Time (ms)
ParaBank Suite			
Regression Tests — passed			
testcases.AccountOverviewTest	testAccountOverviewDetails	1781335561316	1709
testcases.BillPayTest	testBillPayment	1781335573171	5281
testcases.LoginTest	validLogin	1781335547822	1217
testcases.LogoutTest	testLogout	1781335581336	10102
testcases.NegativeLoginTest	testNegativeLogin	1781335551831	1042
testcases.OpenAccountTest	testOpenNewAccount	1781335555672	2792
testcases.RegistrationTest	testRegistration	1781335541447	3204
testcases.TransferFundsTest	testTransferFunds	1781335565985	4168

Regression Tests

testcases.AccountOverviewTest#testAccountOverviewDetails

Parameter #1	Parameter #2
testnguser04	admin123

[back to summary](#)

testcases.BillPayTest#testBillPayment

Parameter #1	Parameter #2
testnguser04	admin123

[back to summary](#)

testcases.LoginTest#validLogin

Parameter #1	Parameter #2
testnguser04	admin123

[back to summary](#)

testcases.LogoutTest#testLogout

[back to summary](#)

testcases.NegativeLoginTest#testNegativeLogin

Parameter #1	Parameter #2
invaliduser	invalidpass

[back to summary](#)

testcases.OpenAccountTest#testOpenNewAccount

Parameter #1	Parameter #2
testnguser04	admin123

[back to summary](#)

testcases.RegistrationTest#testRegistration

Parameter #1	Parameter #2	Parameter #3	Parameter #4	Parameter #5	Parameter #6	Parameter #7	Parameter #8	Parameter #9	Parameter #10	Parameter #11
TestNG 04	User 04	3/12 Main Street	Kolkata	Bengal	700060.0	5.46879213E9	987-123	testnguser04	admin123	admin123

[back to summary](#)

testcases.TransferFundsTest#testTransferFunds

Parameter #1	Parameter #2
testnguser04	admin123

III. Parabank suite report

i. Regression Test

Regression Tests

Tests passed/Failed/Skipped:	8/0/0
Started on:	Sat Jun 13 12:55:36 IST 2026
Total time:	55 seconds (55630 ms)
Included groups:	
Excluded groups:	

(Hover the method name to see the test class name)

PASSED TESTS				
Test method	Attribute(s)	Exception	Time (seconds)	Instance
testRegistration Test class: testcases.RegistrationTest Parameters: TestNG 04, User 04, 3/12 Main Street, Kolkata, Bengal, 700060 0, 5 46879213E9, 987-123, testnguser04, admin123, admin123		3	testcases.RegistrationTest@1ec24091	
testAccountOverviewDetails Test class: testcases.AccountOverviewTest Parameters: testnguser04, admin123		1	testcases.AccountOverviewTest@bccc369	
testNegativeLogin Test class: testcases.NegativeLoginTest Parameters: invaliduser, invalidpass		1	testcases.NegativeLoginTest@13b6aacc	
testTransferFunds Test class: testcases.TransferFundsTest Parameters: testnguser04, admin123		4	testcases.TransferFundsTest@609cd4d8	
testBillPayment Test class: testcases.BillPayTest Parameters: testnguser04, admin123		5	testcases.BillPayTest@17fcd29	
testOpenNewAccount Test class: testcases.OpenAccountTest Parameters: testnguser04, admin123		2	testcases.OpenAccountTest@66d18979	
validLogin Test class: testcases.LoginTest Parameters: testnguser04, admin123		1	testcases.LoginTest@6913c1fb	
testLogout Test class: testcases.LogoutTest		10	testcases.LogoutTest@7d8704ef	

ii. Groups for Parabank Suite

1 suite

Switch Retro Theme

Test results

All suites

ParaBank Suite

Info

Results

Groups for ParaBank Suite

AccountOverviewGroup

testAccountOverviewDetails

BillPaymentGroup

testBillPayment

LoginGroup

validLogin

LogoutGroup

testLogout

OpenAccountGroup

testOpenNewAccount

TransferFundsGroup

testTransferFunds

iii. Times for Parabank Suite

Times for ParaBank Suite				
Total running time: 28 seconds				
Number	Method	Class	Time (ms)	
0	testLogout	testcases.LogoutTest	10,081	
1	testBillPayment	testcases.BillPayTest	5,784	
2	testTransferFunds	testcases.TransferFundsTest	4,043	
3	testRegistration	testcases.RegistrationTest	2,710	
4	testOpenNewAccount	testcases.OpenAccountTest	2,405	
5	testAccountOverviewDetails	testcases.AccountOverviewTest	1,855	
6	validLogin	testcases.LoginTest	1,157	
7	testNegativeLogin	testcases.NegativeLoginTest	849	

iv. Methods in chronological order

Methods in chronological order		
testcases.RegistrationTest		
setup		0 ms
testRegistration(TestNG 12, User 12, 3/12 Main Street, Kolkata, Bengal, 700060.0, 5.46879213E9, 987-123, testnguser12...)		3347 ms
tearDown		6061 ms
testcases.LoginTest		
setup		6415 ms
validLogin(testnguser12, admin123)		8887 ms
tearDown		10045 ms
testcases.NegativeLoginTest		
setup		10354 ms
testNegativeLogin(invaliduser, invalidpass)		12713 ms
tearDown		13564 ms
testcases.OpenAccountTest		
setup		13856 ms
testOpenNewAccount(testnguser12, admin123)		16692 ms
tearDown		19099 ms
testcases.AccountOverviewTest		
setup		19482 ms
testAccountOverviewDetails(testnguser12, admin123)		22507 ms
tearDown		24364 ms
testcases.TransferFundsTest		
setup		24721 ms
testTransferFunds(testnguser12, admin123)		28074 ms
tearDown		32119 ms
testcases.BillPayTest		
setup		32480 ms
testBillPayment(testnguser12, admin123)		35535 ms
tearDown		41321 ms
testcases.LogoutTest		
setup		41639 ms
testLogout		44812 ms
tearDown		54896 ms

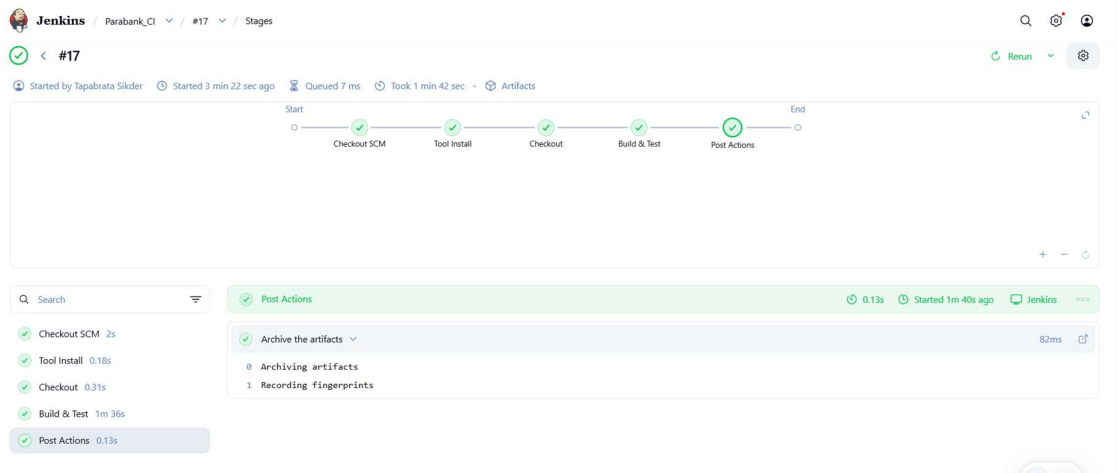
v. Result

1 suite			Test results
All suites			
ParaBank Suite			
Info			
C:\Users\TAPABRATA\git\capstone_project\Parabank\			
1 test			
6 groups			
Times			
Reporter output			
Ignored methods			
Chronological view			
Results			
8 methods, 8 passed			
Passed methods (hide)			
testAccountOverviewDetails(testnguser12, admin123)			
testBillPayment(testnguser12, admin123)			
testNegativeLogin(invaliduser, invalidpass)			
testOpenNewAccount(testnguser12, admin123)			
testRegistration(TestNG 12, User 12, 3/12 Main Street, Kolkata, Bengal, 700060.0, 5.46879213E9, 987-123, testnguser12...)			
testTransferFunds(testnguser12, admin123)			
validLogin(testnguser12, admin123)			
testcases.BillPayTest			
testBillPayment(testnguser12, admin123)			
testcases.TransferFundsTest			
testTransferFunds(testnguser12, admin123)			
testcases.OpenAccountTest			
testOpenNewAccount(testnguser12, admin123)			
testcases.LoginTest			
validLogin(testnguser12, admin123)			
testcases.AccountOverviewTest			
testAccountOverviewDetails(testnguser12, admin123)			
testcases.LogoutTest			
testLogout			
testcases.RegistrationTest			
testRegistration(TestNG 12, User 12, 3/12 Main Street, Kolkata, Bengal, 700060.0, 5.46879213E9, 987-123, testnguser12, admin123, admin123)			
testcases.NegativeLoginTest			
testNegativeLogin(invaliduser, invalidpass)			

12. Jenkins & CI/CD Integration

- **Build Automation:** The project is integrated with Jenkins for Continuous Integration.
- **Workflow:** Every push to the GitHub repository triggers a build in Jenkins, which executes the testng.xml suite via Maven, ensuring immediate feedback on code.

Below is the screenshot of the Jenkins Build output:



```

[INFO] Tests run: 8, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 77.27 s -- in TestSuite
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 8, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 01:32 min
[INFO] Finished at: 2026-06-13T15:56:33+05:30
[INFO] -----

```

- **Branch Strategy & Merge Workflow**

A dedicated feature branch (negative-tests) was created from the master branch to develop and validate negative test scenarios in isolation, following standard Git branching practices. The NegativeLoginTest.java class was developed and verified on this branch without impacting the stable master pipeline. Upon successful local execution and test validation, the negative-tests branch was merged back into master via a pull request. The Jenkins CI/CD pipeline was triggered automatically post-merge, confirming that the full regression suite — including the newly added negative login test — passed without breaking any existing test cases.

13. Risks & Limitations

The following risks and limitations have been identified during the current testing cycle:

- **Application Instability (PAC-3):** The Parabank application throws an Internal Error under concurrent usage conditions. Since the framework currently executes tests sequentially, this has not impacted results. However, if parallel execution is introduced in future, intermittent failures unrelated to the automation code may occur.
- **Test Data Dependency:** The end-to-end flow is tightly coupled — a failure in `RegistrationTest` will cause all downstream tests to fail due to missing session/account data. This architectural risk should be addressed with independent test data setup or `@BeforeClass` data seeding.
- **Public Demo Environment:** The application is a shared public demo. Data created by other users or periodic environment resets can unpredictably affect test execution, particularly for account overview and balance validation steps.
- **Hard-coded Account IDs:** Some test steps reference specific account IDs generated during a particular test run. These will fail on a fresh environment and should be replaced with dynamically captured values.
- **Incomplete Negative Coverage:** Only login negative scenarios have been implemented. Validation defects (PAC-1, PAC-4) may not be caught by regression runs until the remaining negative test cases are developed.

14. Defects Identified During Testing (Defect reporting in JIRA)

Testing identified several functional and stability defects within the Parabank application. These issues have been tracked and logged in Jira for resolution.

Bug ID	Module	Description	Severity
PAC-1	Registration	Lack of input validation on user registration form	High
PAC-2	Transfer	System allows fund transfer to the same account	Medium
PAC-3	General	Application throws Internal Error during concurrent usage	High
PAC-4	Fund Transfer	System accepts negative values in Amount field	High

PAC-5	Update Profile	Phone number displayed in scientific notation	High
PAC-6	Account Overview	Go button on Account Details page is non-functional	High

Spaces

Parabank_Automation_Capstone ...

Summary List Board **Backlog** Development Forms Timeline Docs Reports +

Search backlog Filter

▼ PAC Sprint 1 14 Jun – 28 Jun (6 work items) 0 0 0 Complete sprint ...

☐	PAC-3 [General/Stability]: Application throws "Internal error" during concurrent u...	IN PROGRESS	- = 15
☐	PAC-1 [Registration Module]: Lack of Input Validation on User Registration Form	IN PROGRESS	Jun 20 - ^ 15
☐	PAC-2 [Transfer Module]: System allows fund transfer to the same account	IN PROGRESS	Jun 20 - = 15
☐	PAC-4 [Fund Transfer Module]: System accepts negative values in "Amount" field.	IN PROGRESS	- = 15
☐	PAC-5 [Update Profile Module]: Phone number displayed in scientific notation (E-...	IN PROGRESS	- v 15
☐	PAC-6 [Account Overview]: "Go" button on Account Details page is non-functional.	IN PROGRESS	- ^ 15

Bug ID	Module	Description	Severity
PAC-1	Registration	Lack of input validation on user registration form	High
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PAC-3	General	Application throws "Internal error" during concurrent usage	High
PAC-4	Fund Transfer	System accepts negative values in "Amount" field	High
PAC-5	Update Profile	Phone number displayed in scientific notation	High
PAC-6	Account Overview	"Go" button on Account Details page is non-functional	High

Below are the screenshots for each bug in details:

- **PAC-1:**

6/11/26, 11:13 AM [PAC-1] [Registration Module]: Lack of Input Validation on User Registration Form

[PAC-1] [Registration Module]: Lack of Input Validation on User Registration Form Created: 08/Jun/26 Updated: 08/Jun/26 Due: 20/Jun/26

Status:	To Do
Project:	Parabank_Automation_Capstone
Components:	None
Affects versions:	None
Fix versions:	None

Type:	Bug	Priority:	High
Reporter:	TAPABRATA SIKDER	Assignee:	Unassigned
Resolution:	Unresolved	Votes:	0
Labels:	None		
Remaining Estimate:	Not Specified		
Time Spent:	Not Specified		
Original estimate:	Not Specified		

Attachments:	Screenshot 2026-06-08 121128.png image-20260608-064030.png
Rank:	0j00067:
Start date:	08/Jun/26
Sprint:	PAC Sprint 1

Description

- **Pre-conditions:** The user is on the Parabank Registration page (<https://parabank.parasoft.com/parabank/register.htm>).
- **Steps to Reproduce:**
 1. Navigate to the Registration page.
 2. Enter invalid/special characters into mandatory fields (First Name, Last Name, State, Pincode, Phone, SSN).
 3. Enter a simple numeric password (e.g., "123456").
 4. Click the "Register" button.
- **Expected Result:** The system should display validation errors and reject invalid inputs (e.g., special characters in names, non-numeric values in Pincode/Phone).
- **Actual Result:** The system accepts all invalid inputs and allows the user to proceed with registration despite improper data formats.

```
First name - test10@ -- accepted
Last Name - user10- -- accepted
Street - abc road
City - abxys
State - 20geng -- accepted
Pincode - asd80@ -- accepted
Phone - asd456897 -- accepted
SSN - asd-598
Username - test@10
Password - 123456 -- accepted
```

Generated at Thu Jun 11 05:41:21 UTC 2026 by TAPABRATA SIKDER using Jira 1001.0.0-SNAPSHOT#100291-
rev:2bc9656aea7910482196643967ec7052f5440ae8.

- PAC-2:

6/11/26, 12:20 PM

[#PAC-2] [Transfer Module]: System allows fund transfer to the same account

[PAC-2] [Transfer Module]: System allows fund transfer to the same account Created: 08/Jun/26 Updated: 11/Jun/26 Due: 20/Jun/26

Status:	To Do
Project:	Parabank_Automation_Capstone
Components:	None
Affects versions:	None
Fix versions:	None

Type:	Bug	Priority:	Medium
Reporter:	TAPABRATA SIKDER	Assignee:	Unassigned
Resolution:	Unresolved	Votes:	0
Labels:	None		
Remaining Estimate:	Not Specified		
Time Spent:	Not Specified		
Original estimate:	Not Specified		

Attachments:	 Screenshot 2026-06-08 121128.png
Rank:	0 j0006f:
Start date:	08/Jun/26
Sprint:	PAC Sprint 1

Description

- **Pre-conditions:** The user is logged into their Parabank account, and at least one account exists with a positive balance.
- **Steps to Reproduce:**
 1. Navigate to the "Transfer Funds" page.
 2. In the "From Account" dropdown, select Account X.
 3. In the "To Account" dropdown, select the same Account X.
 4. Enter a valid amount (e.g., 100).
 5. Click the "Transfer" button.
- **Expected Result:** The application should display an error message stating that transfers to the same account are not permitted.
- **Actual Result:** The system processes the transaction, shows a "Transfer Complete" message, and modifies the balance incorrectly.
- **Severity:** High



Generated at Thu Jun 11 06:48:42 UTC 2026 by TAPABRATA SIKDER using Jira 1001.0.0-SNAPSHOT#100291-rev:2bc9656aea7910482196643967ec7052f5440ae8.

- PAC-3:

6/11/26, 12:22 PM

[#PAC-3] [General/Stability]: Application throws "Internal error" during concurrent user activity

[PAC-3] [General/Stability]: Application throws "Internal error" during concurrent user activity Created: 08/Jun/26 Updated: 11/Jun/26

Status:	To Do
Project:	Parabank_Automation_Capstone
Components:	None
Affects versions:	None
Fix versions:	None

Type:	Bug	Priority:	Medium
Reporter:	TAPABRATA SIKDER	Assignee:	Unassigned
Resolution:	Unresolved	Votes:	0
Labels:	None		
Remaining Estimate:	Not Specified		
Time Spent:	Not Specified		
Original estimate:	Not Specified		

Attachments:	 image-20260608-101933.png
Rank:	0 j00063:
Sprint:	PAC Sprint 1

Description

- **Pre-conditions:** Multiple users are performing transactions on the Parabank demo site simultaneously.
- **Steps to Reproduce:**
 1. Log in to the application as a registered user.
 2. Navigate to a service page (e.g., "Transfer Funds" or "Account Overview").
 3. Attempt to perform an action (e.g., Transfer) while others are also using the site.
- **Expected Result:** The application should process the request successfully or provide a user-friendly error message if the system is overloaded.
- **Actual Result:** The application displays the message: "An internal error has occurred and has been logged."
- **Severity: High** (Because it blocks core functionality).



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- PAC-4:

6/11/26, 12:22 PM

[#PAC-4] [Fund Transfer Module]: System accepts negative values in "Amount" field.

[PAC-4] [Fund Transfer Module]: System accepts negative values in "Amount" field. Created: 08/Jun/26 Updated: 11/Jun/26			
Status:	To Do		
Project:	Parabank_Automation_Capstone		
Components:	None		
Affects versions:	None		
Fix versions:	None		
Type:	Bug	Priority:	Medium
Reporter:	TAPABRATA SIKDER	Assignee:	TAPABRATA SIKDER
Resolution:	Unresolved	Votes:	0
Labels:	None		
Remaining Estimate:	Not Specified		
Time Spent:	Not Specified		
Original estimate:	Not Specified		
Attachments:	 image-20260608-155227.png		
Rank:	0 0006n:		
Sprint:	PAC Sprint 1		

Description	
• Pre-conditions: The user is logged into Parabank and has accounts with positive balances. • Steps to Reproduce: 1. Navigate to the "Transfer Funds" page. 2. Enter a negative value (e.g., -20) in the "Amount" field. 3. Select a valid "From" and "To" account. 4. Click the "Transfer" button. • Expected Result: The system should reject the negative value and display an error message (e.g., "Amount must be a positive number"). • Actual Result: The system processes the transaction, and the "Transfer Complete" message is displayed, indicating a successful transfer of a negative amount. • Attachments: Attach both the screenshot showing the input field with -20 () and the resulting confirmation page (). Transfer Complete! \$20.00 has been transferred from account #15000 to account #14000 See Account Activity for more details.	

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- PAC-5

6/11/26, 2:16 PM

[#PAC-5] [Update Profile Module]: Phone number displayed in scientific notation (E-format) on Update Profile page.

[PAC-5] [Update Profile Module]: Phone number displayed in scientific notation (E-format) on Update Profile page. Created: 08/Jun/26 Updated: 08/Jun/26

Status:	To Do
Project:	Parabank_Automation_Capstone
Components:	None
Affects versions:	None
Fix versions:	None

Type:	Bug	Priority:	Low
Reporter:	TAPABRATA SIKDER	Assignee:	Unassigned
Resolution:	Unresolved	Votes:	0
Labels:	None		
Remaining Estimate:	Not Specified		
Time Spent:	Not Specified		
Original estimate:	Not Specified		

Attachments:	 Screenshot 2026-06-08 212415.png
Rank:	0 j0006v:
Sprint:	PAC Sprint 1

Description

- - **Pre-conditions:** The user has a profile with a 10-digit phone number.
 - **Steps to Reproduce:**
 1. Log in to the Parabank application.
 2. Navigate to the "Update Contact Info" page.
 3. Observe the "Phone #" field.
 - **Expected Result:** The phone number should be displayed exactly as it was entered (e.g., 5468792139).
 - **Actual Result:** The phone number is displayed in exponential/scientific notation format: 5.46879213E9.
- **Severity: Medium** (It is a UI/UX and data retrieval defect, though it does not crash the system).

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- **PAC-6:**

6/11/26, 2:17 PM

[#PAC-6] [Account Overview]: "Go" button on Account Details page is non-functional.

[PAC-6] [Account Overview]: "Go" button on Account Details page is non-functional. Created: 08/Jun/26 Updated: 08/Jun/26

Status:	To Do
Project:	Parabank_Automation_Capstone
Components:	None
Affects versions:	None
Fix versions:	None

Type:	Bug	Priority:	High
Reporter:	TAPABRATA SIKDER	Assignee:	Unassigned
Resolution:	Unresolved	Votes:	0
Labels:	None		
Remaining Estimate:	Not Specified		
Time Spent:	Not Specified		
Original estimate:	Not Specified		

Attachments:	 Screenshot 2026-06-08 213445.png
Rank:	0 00073:
Sprint:	PAC Sprint 1

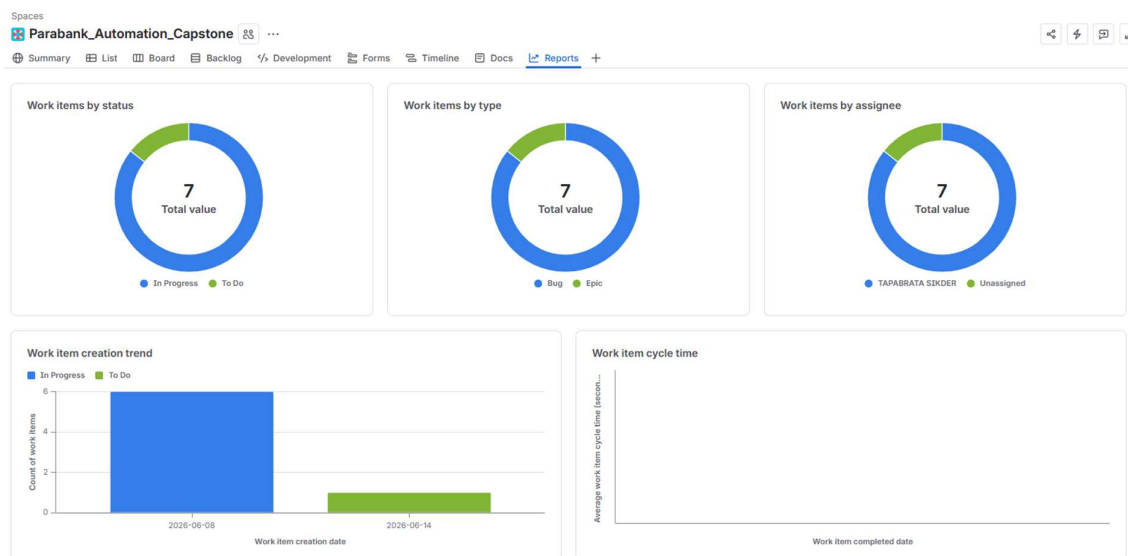
Description

- **Pre-conditions:** The user is logged in and has one or more accounts listed on the "Accounts Overview" page.
- **Steps to Reproduce:**
 1. Navigate to the "Accounts Overview" page.
 2. Locate the account details section and the "Go" button next to the account selection/details area.
 3. Click the "Go" button.
- **Expected Result:** The system should navigate the user to the transaction history or details page for the selected account.
- **Actual Result:** No action is performed, the page does not refresh, and no navigation occurs.

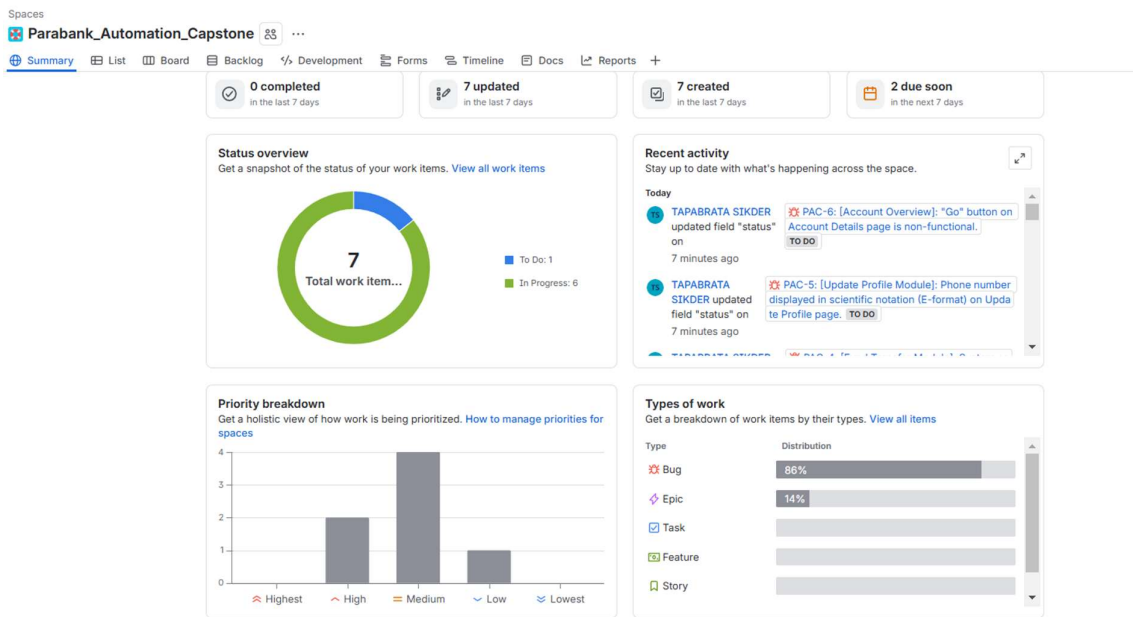
Severity: High.

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Reports



Summary



15. Conclusion

The Para Bank Automation Test Plan demonstrates a well-structured, scalable, and maintainable test automation framework built on industry-standard tools and design principles. The Page Object Model architecture ensures clear separation of concerns, while data-driven testing via Apache POI enables flexible and reusable test scenarios. The integration of Jenkins CI/CD with GitHub ensures that every code change is validated automatically, providing rapid feedback and preventing regression.

In the current execution cycle, all 8 test cases — including one negative login scenario — achieved a 100% pass rate. The introduction of a dedicated feature branch (negative-tests) and its successful merge into master demonstrates adherence to professional Git branching workflows and reflects a real-world SDET delivery process.

Six functional defects of High and Medium severity were identified and logged in JIRA, providing clear traceability between test execution and application quality issues. The Requirements Traceability Matrix ensures that all in-scope features are covered and linked to their respective test cases and defect reports.

Key areas identified for future improvement include: expanding negative test coverage to include duplicate registration, empty form submissions, and invalid fund transfer amounts;

replacing hard-coded account IDs with dynamically captured values; and introducing a TestNG RetryAnalyzer to handle potential flakiness in CI environments. Addressing these gaps will further strengthen the framework's reliability, coverage, and suitability for enterprise-level regression testing.

Summary Metric	Value
Total Test Cases	8
Pass Rate	100%
Negative Tests Implemented	(NegativeLoginTest)
Defects Logged in JIRA	6
CI/CD Pipeline	Jenkins + GitHub (Active)
Branching Strategy	Feature branch workflow (merged to master)